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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Attempt any FIVE of the following : 10

(a) State the general principles of preservation.

(b) Define TDT, D-value, Z-value, F-value.

(c) State preservation principle of low temperature.

(d) Differentiate between slow and quick freezing methods.

(e) Define drying & Dehydration.

(f) State the principle of pulsated Electrical Field.

(g) State the principle HPP and MATS.

2. Attempt any THREE of the following : 12

(a) List the methods of food preservation and state major preservation factor.

(b) Define blanching process and list the processing parameters of different foods.

(c) State the factors affecting chilling and frozen storage of foods.

(d) Explain principle of preservation of Ultrasound processing.

3. Attempt any THREE of the following : 12

(a) Outline the effect of heat processing on quality of food.

(b) Describe factors affecting TDT.

(c) State name and permissible level of addition of preservative used in preparation of ketchup as per FSSAI Act, 2006.

(d) Explain the mechanism of sulphur dioxide as preservatives.

4. Attempt any THREE of the following : 12

(a) Classify food on the basis of ease of spoilage.

(b) Explain the factors affecting heat penetration.

(c) Explain working of equipment's used for chilling and freezing of foods. (any two equipments)

(d) State the Principle of preservation of Salt & Sugar.

(e) State the Irradiation dose level limits used for inhibiting sprouting in Tubers and disinfestation of dry Spices as per FSSAI Act, 2006.

5. Attempt any TWO of the following : 12

(a) Describe the canning process with flow diagram.

(b) Explain the freezing curve and the changes occur during thawing of foods.

(c) Outline the principle preservation of Hurdle technology with suitable example.

6. Attempt any TWO of the following : 12

(a) Outline the growth curve of microorganisms.

(b) Describe the following preservation techniques :

(i) Pasteurization

(ii) Sterilization

(c) Describe the drying rate curve with diagram.